(Conservation of momentum)

2. $\frac{1}{2}M_1u_1+\frac{1}{2}m_2u_2 = m_1v_1^2+\frac{1}{2}m_2v_2^2$
(Conservation of kinetic energy)

Nature of Elastic Collision

1. The bodies involved always separate after the collision that is they move with different velocities after collision.
2. The bodies involved in this type of collision do not undergo any deformation after the collision.

Inelastic Collision

A collision is said to be inelastic when only momentum is conserved but kinetic energy is not conserved in the event of the collision.

The equations describing the type of collision are

1. $M_1u_1+m_2u_2 = (m_1+m_2)v$

After collision the two bodies stick together and move with common velocity. wherev is the common velocity with which the two bodies move.

Nature of Inelastic Collision

1. Due to energy transformations, kinetic energy before collision is greater than kinetic energy after collision.
2. Bodies involved in this type of collision stick together after collision and move with a common velocity.

Solved Examples

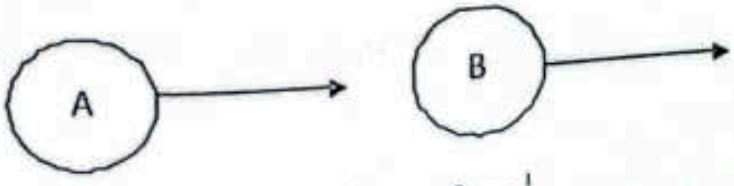
1. A body of mass 60kg moving with velocity of 5m/s hits a stationary body of mass 30kg.

a) If the bodies move together after the impact, calculate the velocity of the composite body just after the impact.

b) If the bodies do not move together and the body of mass 30kg is moving with a velocity of 2 m/s in the same direction as the body of mass 60kg, find the velocity of the 60kg mass after impact.

Solution

a) Before impact



M1 – 60kg; m, – 30kg u1 – 5ms^{1}; u, – 0ms^{1}

Momentum before impact = $M_1u_1+m_2u_2$

Momentum before impact = $60 \times 5 + 30 \times 0 = 300$kg/ms
Momentum after impact



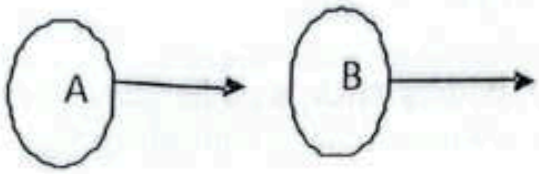
$m_1 = 60kg$; $m_2 = 30kg$; $v = ?$

Momentum after impact = $(m_1 + m_2)v = (60 + 30)v = 90v$
From the principle of conservation of linear momentum

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

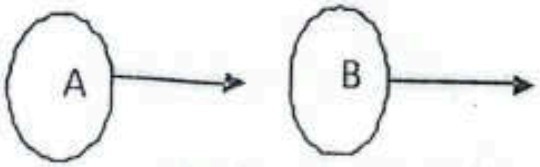
$$300 = 90v \; ; \; v = \frac{300}{90} = 3.33m/s$$

B) Before impact



$m_1 = 60kg$; $m_2 = 30kg$

$u_1 = 5ms^{-1}$; $u_2 = 0ms^{-1}$



$m_1 = 60kg$; $m_2 = 30kg$

$u_1 = ?$; $u_2 = 2ms^{-1}$

Momentum after impact = $m_1v_1 + m_2v_2$
$= 60u_1 + 30 \times 2$
$= 60u_1 + 60$
$300 = 60u_1 + 60$

$$u_1 = \frac{240}{60} = 2m/s$$

2.a) What is the difference between *inertia* and *momentum*?

B) A bullet of mass 25.0g is fired with a velocity of 50.0m/s from a gun

of mass 2.5kg. calculate the recoil velocity of the gun

Solution

a) Inertia is the tendency of a body to remain at rest or continue its motion in a straight line unless it is acted upon by an external force to change its state. **Momentum** on the other is the product of mass and velocity. **inertia** is scalar quantity and momentum is vector quantity. **inertia** involves mass of a body whilst **momentum** involves mass and velocity of a body.

b) Let the mass of the bullet be $m_b = 25 \times 10^{-3}$kg
Velocity of the bullet, $u_b = 50$m/s
Mass of gun, $m_g = 2.5$kg,
Recoil velocity of gun $= u_g = ?$
From law of conservation of momentum

$$m_g u_g = m_b u_b \text{ and } u_g = \frac{m_b u_b}{m_g}$$

$$u_g = \frac{25\times10^{-3}\times50}{2.5} = 0.5\text{m/s}$$

4.2.2 Weightlessness Motion in a lift

Everyday activities in transporting objects in a skyscraper involves a moving device called a lift, and the working principle of this partially works with cranes, when the lift accelerates downwards occupants feel lighter, however they feel heavier when it accelerates upwards. These changes can be explained by the resulting force from ***the force of gravity*** and ***the force needed to accelerate or decelerate the lift.***

Weight equation for motion in lift

If the lift accelerates downward **(W = mg - ma)**

If the lift accelerates upwards **(W = mg + ma)**

Solved Examples

1. A man of mass 60kg is standing in a lift. What force does the floor of the lift exert on the man if the lift is (take $g = 9.8\text{m/s}^2$)

i) Moving with a uniform velocity?

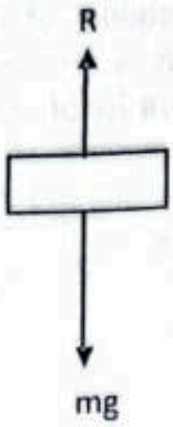
ii) Accelerating at 3m/s^2 upwards?

iii) Accelerating at 3m/s^2 downwards?

Solution

R is the normal reaction from the floor on the man

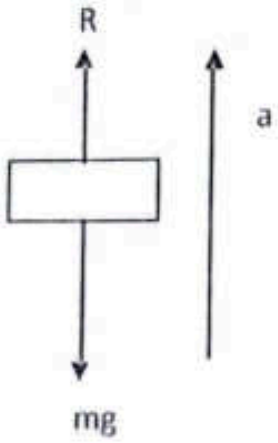
I)



Since the lift is moving with a uniform velocity, the resulting force is R = mg

$$R = 60 \times 9.8 = 588N$$

ii)

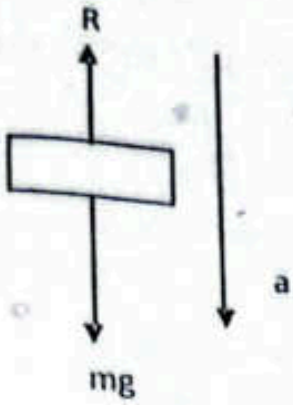


Since the lift is accelerating upwards

Equation of motion:

$$R = mg = ma$$
$$R = m(g + a)$$
$$R = 60(9.8 + 3)$$
$$R = 768N$$

(ii)



Since the lift is accelerating upwards
Equation of motion:

$$mg - R = ma$$
$$R = m(g - a)$$
$$R = 60(9.8 - 3)$$
$$R = 408N$$

4.3 Connected Bodies (Atwood Machine)

Two particles connected by a light inextensible string passing over a fixed light smooth frictionless pulley are called connected bodies. The tension in the string is the same throughout its length so the body is acted upon by the same tension. Problems concerned with the connected bodies usually involve finding the acceleration of the system and the tension in the string.

Cases of connected bodies in motion

1. Particle on a smooth or rough horizontal table

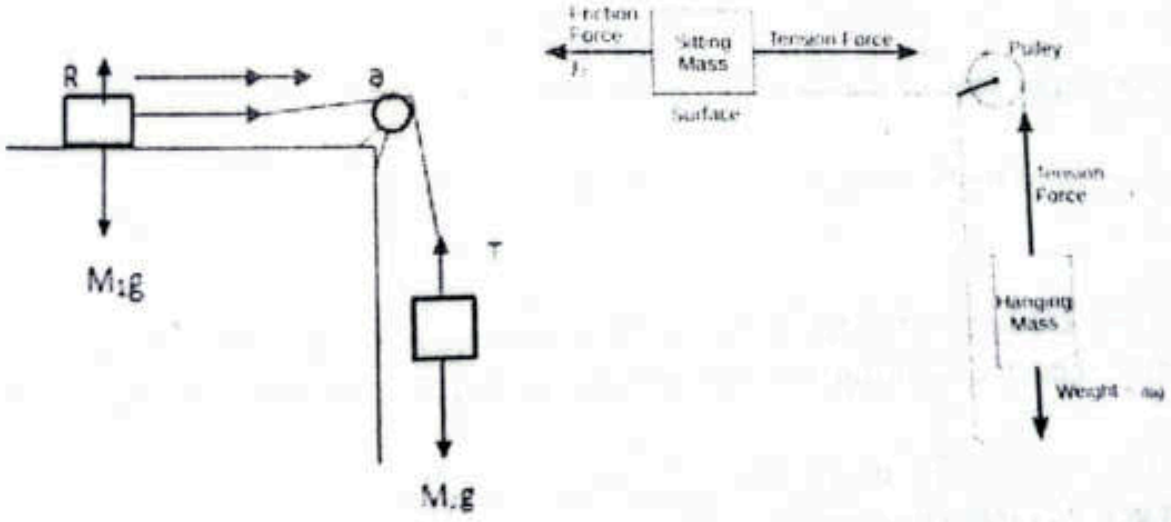


Figure 4.1: suspended bodies on a pulley

Smooth surface table

For m_1

Equation of motion: $T = m_1 a$

For m_2

Equation of motion: $m_2 g \;\; T = m_2 a$

Rough surface table

For m_1

Equation of motion: $1 \;\; F = m_1 a$

For m_2

Equation of motion: $m_2 g \;\; T = m_2 a$

1. Atwood machine

Atwood machine was invented in 1784 by the English mathematician George Atwood as a laboratory experiment to verify the mechanical laws of motion with constant acceleration. it consists of two objects of masses m_1 and m_2 connected by an inextensible massless string over an ideal massless pulley

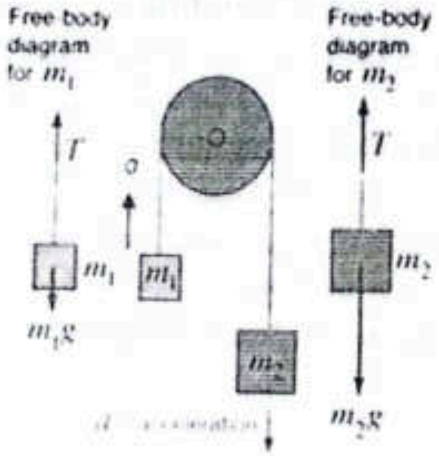


Figure 4.2: Atwood machine

Equation of motion for $m_1 \Rightarrow F_{net} = T - m_1 g$

Equation of motion for $m_2 \Rightarrow F_{net} = m_2 g - T = m_2 a$

For this idealized case the tension T, is the same on both sides of the pulley. The acceleration a is the same for both masses, Solving for T gives

$$T = m_1 g + m_2 a \quad \text{.....(1)}$$

Put eqn. (1) into equation m_2

$$m_2 g - T = m_2 a \quad \text{.....(2)}$$

$$m_2 g - (m_1 g + m_1 a) - m_2 a$$

$$m_2g - m_1g - m_1a = m_2a$$
$$m_2g - m_1g = m_2a + m_1a$$

Rearranging the equation, we get the acceleration to be

$$a = \frac{(m_2 - m_1)g}{(m_2 + m_1)}$$

The acceleration equation changes when mass m_1 moves down and m_2 moves up, the sign convention changes.

$$a = \frac{(m_1 - m_2)g}{(m_1 + m_2)}$$

Putting the acceleration equation into eqn. (1)

$$T = m_1(g + a)$$

$$= m_1\left[g + \left[\frac{m_2 - m_1}{m_1 + m_2}\right]g\right]$$

$$= m_1g\left[1 + \frac{m_2 - m_1}{m_1 + m_2}\right]$$

Rearranging the equation, we can get the tension T, to be

$$T = \left[\frac{m_1m_1}{m_1 + m_2}\right]g$$

Solved examples

1. Two bodies of masses 5 kg and 3 kg are connected by a light string which passes over a fixed light smooth frictionless pulley. Find the common acceleration a and the tension T in the string when the system is moving freely(take g = 10m/s)

Solution

since m_1 is heavier than m_2 then m_1 moves down while m_2 moves up in the pulley system.

Using

$$a = \frac{(m_1 - m_2)g}{(m_1 + m_2)}$$

$$= \frac{(5 - 3)10}{(5 + 3)} = 2.5m/s$$

(b). Applying

$$T - \left[\frac{m_1m_2}{m_1 + m_2}\right]g$$

$$= \left[\frac{2 \times 5 \times 3}{(5 + 3)}\right]10 = 37.5N$$

4.4 CENTER OF GRAVITY

The center of gravity of an object is the point through which the total weight of the body can be considered to act.

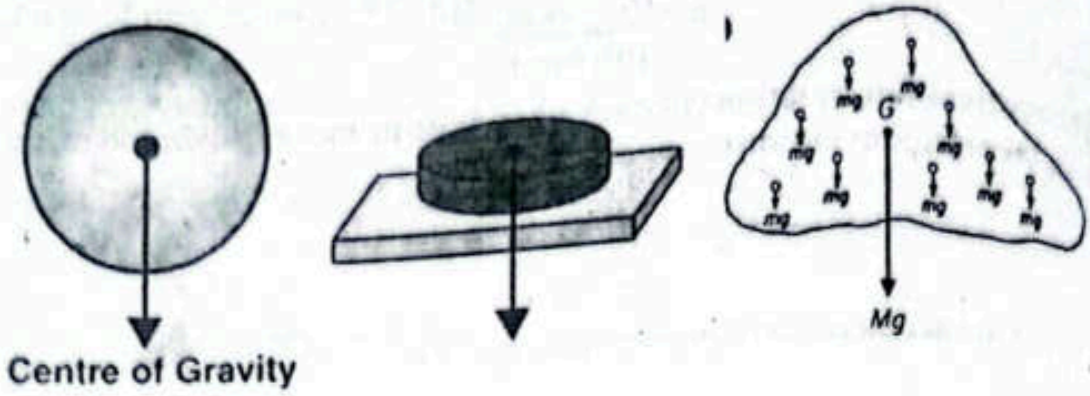


Figure 4.3: Centre of gravity of a body

The weight of a body acts through its center of gravity.

4.4.1 CENTER OF MASS

This is a point where the total mass of a body appears to act. *for small objects the center of mass coincides with the center of gravity.*

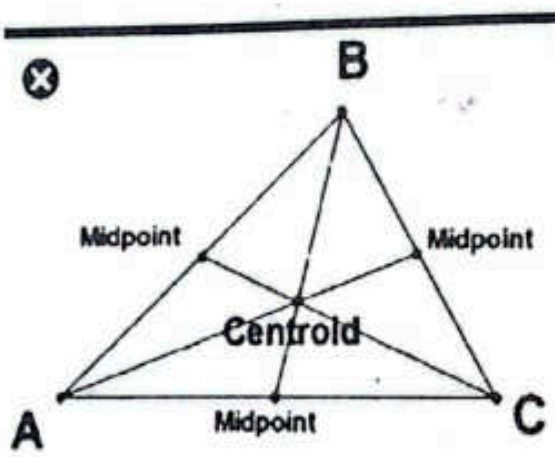


Figure 4.4: Centre of mass of a body

The center of mass of a body or a system of bodies is the point that moves as through all of the masses concentrated there and all external forces where applied there. The position of center of mass of n particles along x-axis is given as

$$Xcon = \frac{m_1x_1 + m_2x_2 + m_3x_3 + \ldots\ldots + m_nx_n}{m}$$

$$\frac{1}{m}\sum_{i=1}^{n} m_i x_i$$

When $m = m_1 + m_2 + m_3 + \ldots + m_n$

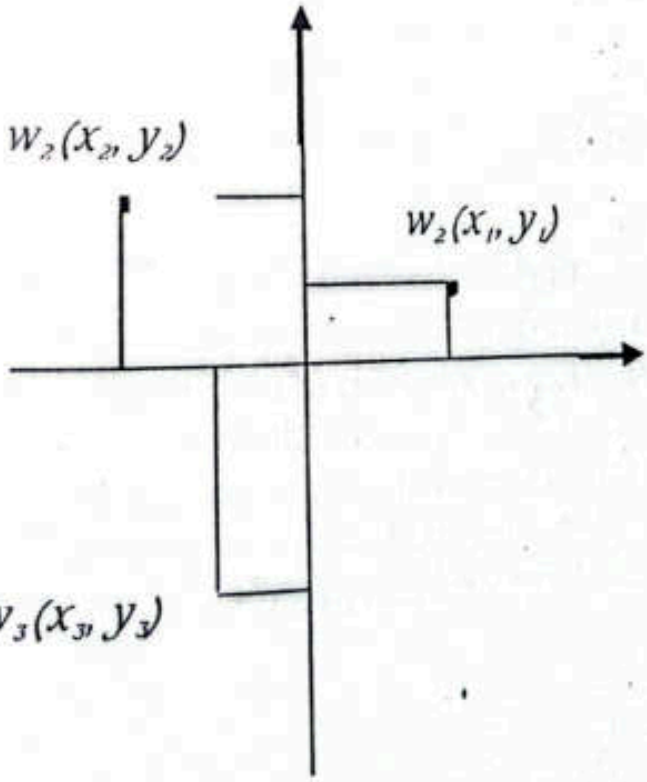
If the particles are distributed in three dimensions, the center mass must be identified by three coordinates

$$X_{com} = \frac{1}{m}\sum_{i=1}^{n} m_i x_i$$

$$Y_{com} = \frac{1}{m}\sum_{i=1}^{n} m_i y_i$$

$$Z_{com} = \frac{1}{m}\sum_{i=1}^{n} m_i z_i$$

Consider the particles with weight W_1, W_2 and W_3 with position (X_1, Y_1), (X_2, Y_2) and (X_3, Y_3) relative to x and y axes in their plane as shown below.



Applying principle of moment,

$$W_1x_1 + W_2x_2 + W_3x_3 = \quad wx$$

$$\bar{X} = \frac{W_1x_1 + W_2x_2 + W_3x_3}{\Sigma W}$$

$$\bar{Y} = \frac{W_1y_1 + W_2y_2 + W_3x_3}{\Sigma W}$$

Center of gravity = $(\bar{X}, \bar{Y})$

Solved examples

1. Three masses 6 kg, 2 kg and 4 kg are located at position (2,3), (-2,2) and (3, -1) respectively on the x-y plane. Determine the center of mass of the system.

Solution

From $$\bar{X} = \frac{W_1x_1 + W_2x_2 + W_3x_3}{\Sigma W}$$

$$\bar{X} = \frac{(60 \times 2) + (20 \times (-2) + (40 \times 3)}{60+20+40}$$

$$= 1.67$$

$$\frac{W_1y_1 + W_2y_2 + W_3x_3}{\Sigma W}$$

$$\bar{Y} = \frac{(60 \times 3) + (20 \times 2) + (40 \times (-1)}{60+20+40}$$

$$= 1.5$$

Center of gravity = $(\bar{X}, \bar{Y})$ = (1.67, 1.5)

4.5 Frictional Force

Friction is a force resisting relative motion and it occurs at the interface between two bodies. In case of fluids it occur within the bodies (i.e. between molecules of fluids). When an object is in motion either on a surface or in a viscous medium such as air or water, there is resistance to the motion because the objects interacts with the surroundings. The resistance is called a "force of friction". The principle helps us to walk and in wear and tear of tires, example is as shown in the figures below:

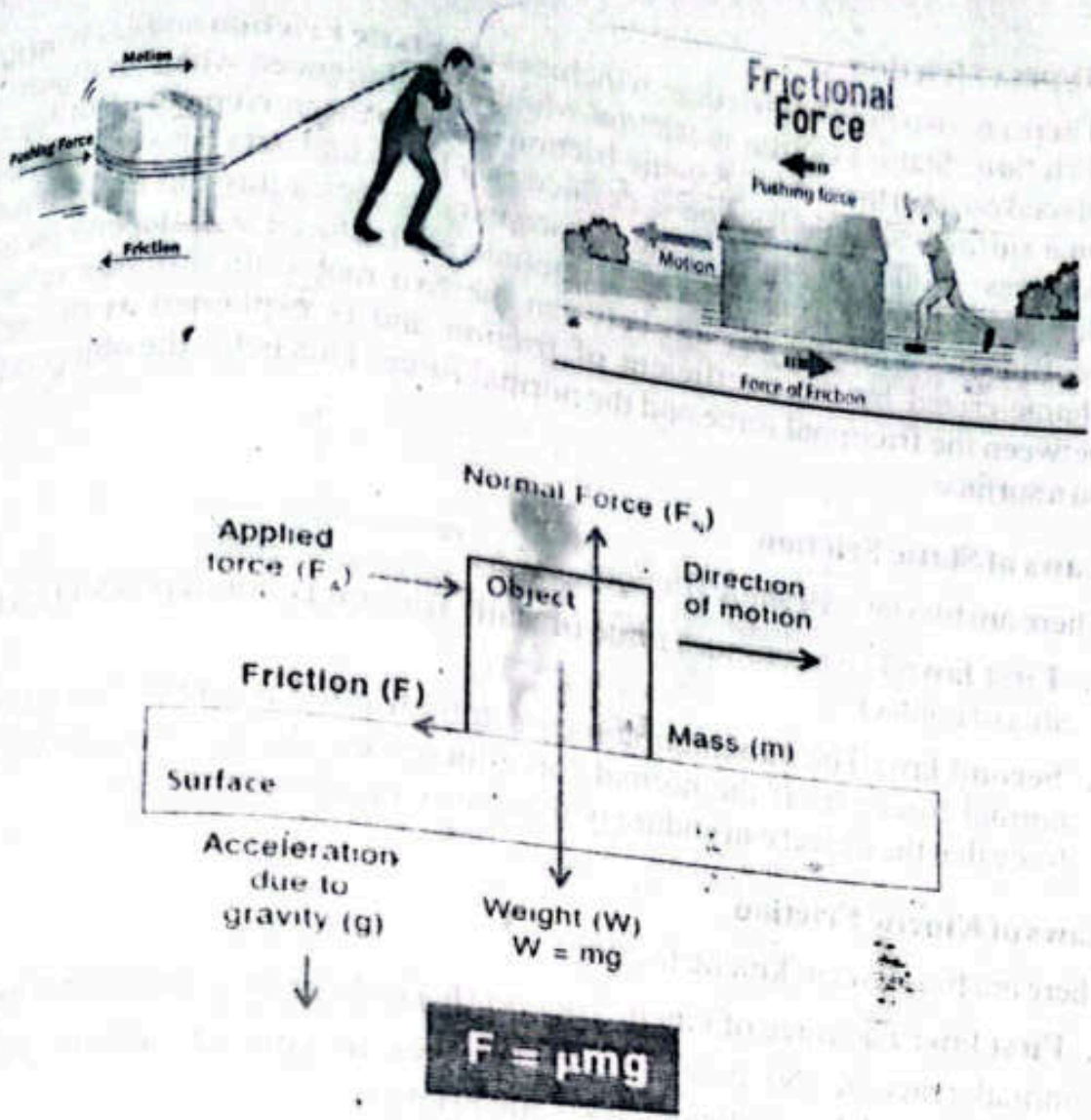


Figure 4.5: Frictional force of a moving body

Laws of Friction

There are five laws of friction and they are:

- The friction of the moving object is proportional and perpendicular to the normal force.
- The friction experienced by the object is dependent on the nature of the surface it is in contact with.
- Friction is independent of the area of contact as long as there is an area of contact.
- Kinetic friction is independent of velocity.
- The coefficient of static friction is greater than the coefficient of kinetic friction.

Types of friction

There are two types of friction; it includes (1) **Static Friction** and (2) **Kinetic Friction**. Static Friction is friction which is experienced when an object is placed on a surface. And, Kinetic friction is due to the movement of an object on a surface. Kinetic friction is defined as a force that acts between moving surfaces. A body moving on the surface experiences a force in the opposite direction of its movement. The magnitude of the force will depend on the coefficient of kinetic friction between the two materials. Friction is well characterized by the coefficient of friction and is explained as the ratio between the frictional force and the normal force. This helps the object to lie on a surface.

Laws of Static Friction

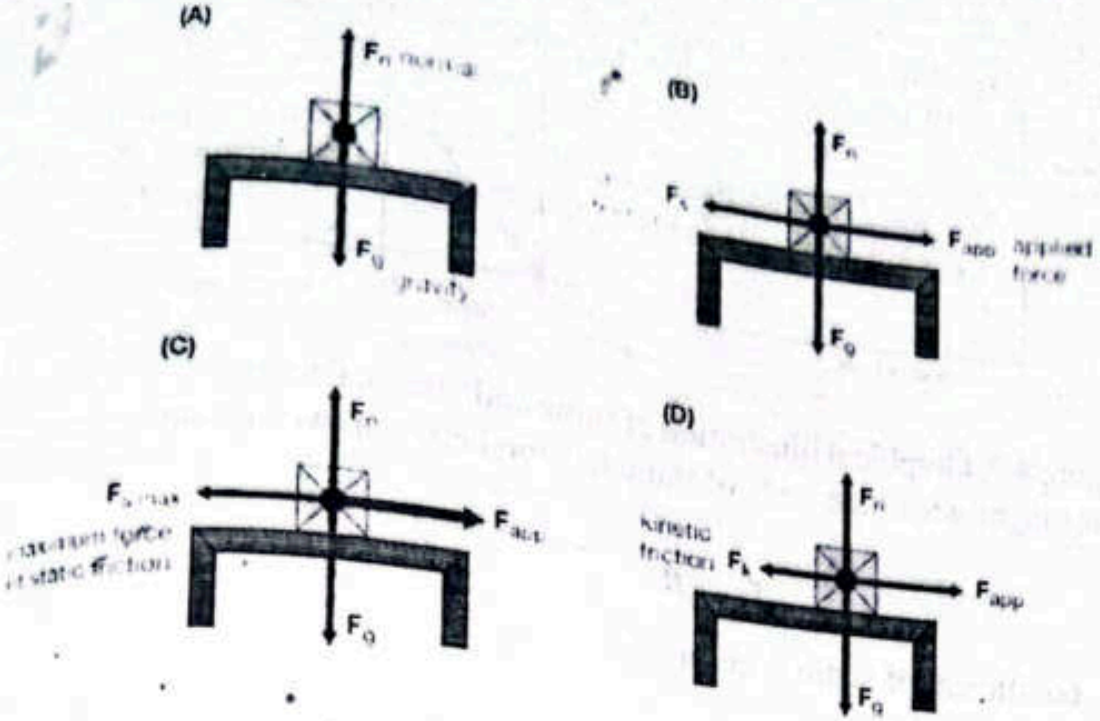
There are two laws of static friction:

1. **First law:** The maximum force of static friction is not dependent on the area of contact.
2. **Second law:** The maximum force of static friction is comparative to the normal force i.e., if the normal force increases, the maximum external force that the object can endure without moving, also increases.

Laws of Kinetic Friction

There are four laws of kinetic friction:

1. **First law:** The force of kinetic friction (F_k) is directly proportional to the normal reaction (N) between two surfaces in contact. Where, μ_k? = constant called the coefficient of kinetic Friction.
2. **Second law:** Force of kinetic friction is independent of shape and apparent area of the surfaces in contact.
3. **Third law:** It depends upon nature and material of surface in contact.
4. **Fourth law:** It is independent of the velocity of object in contact provided the relative velocity between the object and the surface is not too large.



Static Friction

Static friction is the friction present between two or more objects that not moving with respect to each other

The magnitude of static friction is greater due to the greater value of its coefficient.

The equation representing static friction is given by

$F_s = \mu_s \eta$

Its value can be Zero.

Example-A pencil on the table.

Kinetic Friction

Kinetic friction is the friction present between two or more objects that are in motion with respect to each other.

The magnitude of the kinetic friction is comparatively lesser due to the low value of its coefficient.

The equation representing kinetic friction is given by

$F_k = \mu_k \eta$

Its value can never be zero.

Example- Moving the pencil over a table top.

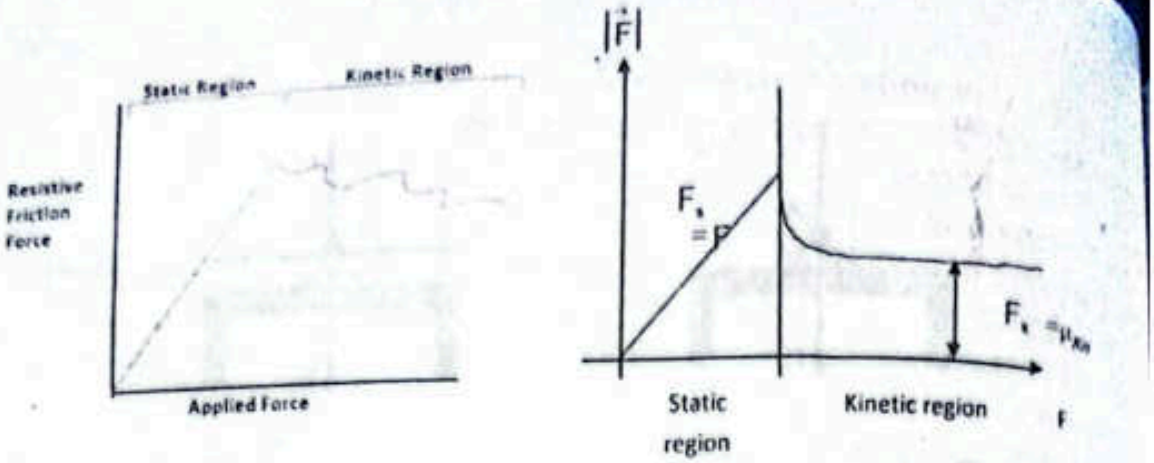


Figure 4.7: Graphical illustration of static and kinetic friction

The magnitude of the force of static friction between two surfaces in contact is:

$$F_s = \mu_s n$$

μ_s = coefficient of static friction

n= normal reaction

The magnitude of the force of kinetic friction acting between the surfaces is:

$$F_s = \mu_k n$$

μ_k = coefficient of kinetic friction

n= normal reaction

μ_s and μ_k = depends on the nature of the surface μ_k is generally less than μ_s

Solved Examples

1. **A worker pushes large cardboard of mass 93.0 kg box across the floor.** The coefficient of kinetic friction is μ_k, what is the magnitude of the force of friction if the worker exerts a force of 500.0 N forward. Determine **the net force moving the box.**

Solution

On a flat surface, the normal force of an object can be found out by the formula

$$n = mg$$

By substituting the value of in the equation

$$F_k = \mu_k n, \text{ we get}$$

$$F_k = (0.520)\ (93.0\text{kg})\ (9.80\text{m/s}^2) = 473.928\text{N}$$

The net force acting on a body is the sum of all the forces acting on the body. In this case, the forces acting on the body are the force exerted by the man and the kinetic friction acting in the opposite direction. If the forward motion is considered positive, then the net force is calculated as follows.

$F_{net} = F - F_k$

Substituting the values in the above equation, we get

F_{net} 500 - 473.928N - 26.072 N

12. An object having a mass of 20 kg is placed on a smooth surface. Static friction between these two surfaces is given as the 35 N. Find the coefficient of static friction?

Solution

Recall thatNormal force, $n = mg$

$$So, n = 20 \times 9.81 = 197N$$

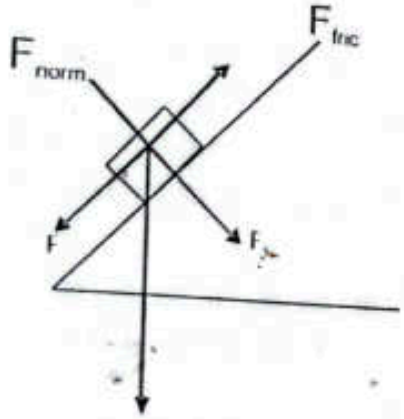
The coefficient of static friction can be determined using:

$$F = \mu_s n$$

$$\mu_s = F_s/n = 35/195$$

3. The forces acting upon a 100kg crate that is sliding down an inclined plane. The plane is inclined at an angle 30. The coefficient of friction between the crate and the incline is 0.3. Determine the net force and acceleration of the crate (take $g = 9.8 m/s^2$).

Solution



Mass = 100 kg, $\mu_k = 0.3$

$$mg = 100 \times 9.8 = 980N$$

$$F = mg \sin 0 = 980 \times \sin 30^0 = 490N$$

$$F_1 = mg \sin \theta = 980 \times \sin 30^0 = 490N$$

F_{norm} = (It must be balance with the perpendicular force)

$$F_{fric} = \mu N = 0.3 \times 849 = 255N$$

$$F_{net} = F_{norm} + F + (-F_{fric})$$

$$= [849 + (-849) + 490 + (255)$$

$$F_{net} = 235N$$

To calculate the acceleration

$$F_{net} = ma$$

$$a = \frac{F_{net}}{m}$$

$$= \frac{235}{100} = 2.35 m/s^2$$

Practice Questions

1. What is the tension in a vertical rope pulling a block of wood of mass 70kg with an acceleration of 2m s^{-2} **(Ans: T = 826N)**
2. A force of magnitude 15N acts on a 4kg body moving initially in the direction of the force with speed of 4m/s. Calculate the distance travelled by the body if the final speed of the body is 6m/s? **(Ans: 2.67m)**
3. A 15kg load is attached to a hanging 25kg mass. Friction is negligible. What is the acceleration of the 15kg cart?**(Ans: 6.1m/s^2)**
4. A man kicks a stationary ball of mass 200g, giving it a speed of 100m/s, what impulse is imparted to the ball?**(Ans: 20Ns)**
5. State the Newton's laws of motion, and indicate the relevance to force field on everyday mechanics.
6. A body of mass 10 kg moving with a velocity of 5m/s collides with a stationary body of mass 4 kg and stick together. They then move with a common velocity V, find the value of V.**(Ans: 3.57m/s)**
7. The normal force and the static frictional force of an object are 50 N and 80 N respectively. Find the coefficient of static friction?(Ans : $\mu_s = 1.6$)
8. A fridge weighs 1619 N and the static friction coefficient is 0.50. What is the least force used to move the fridge?(Ans: F = 809.50 N)
9. A particle of mass 10 kg is placed on a rough plane inclined at an angle of 30 to the horizontal. What force acting parallel to the plane is required
   i) To hold the particle at rest?
   ii) To just make the particle move up the plane? (take m/s)**(Ans: F = 254 N)**
10. A particle of mass 5 kg is placed on a smooth plane inclined at an angle of 30 to the horizontal. What force
    i) acting parallel to the plane
    ii) acting horizontally is required to hold the particle at rest?
11. The coefficient of static friction between a box of mass 15.0kg and the horizontal surface is 0.4. If the force of 20 N is applied to the box, calculate the friction force and the limiting frictional force between the two surfaces.
12. Three masses 6 kg, 2 kg and 4 kg are located at position (2,3), (-2,2) and (3, -1) respectively on the x-y plane. Determine the center of mass of the system.**(Ans: 1.67,1.5)**
13. A car with a mass of 1000kg accelerates from 3m s to 10m s. The

force of the engine acting on the car is 500N. Over what distance did the force act. **(Ans: 91m)**

14. A force of magnitude 5N acts on 2kg body moving initially in the direction of the force with a speed of 4m/s. over what distance must the force act in order to change the body's speed to 6m/s? **(Ans: 4m)**

15. A car of mass 1000kg is accelerated from rest to a speed of 30m/s in 20sec. what is the average force applied to the car. **(Ans = 1500N)**